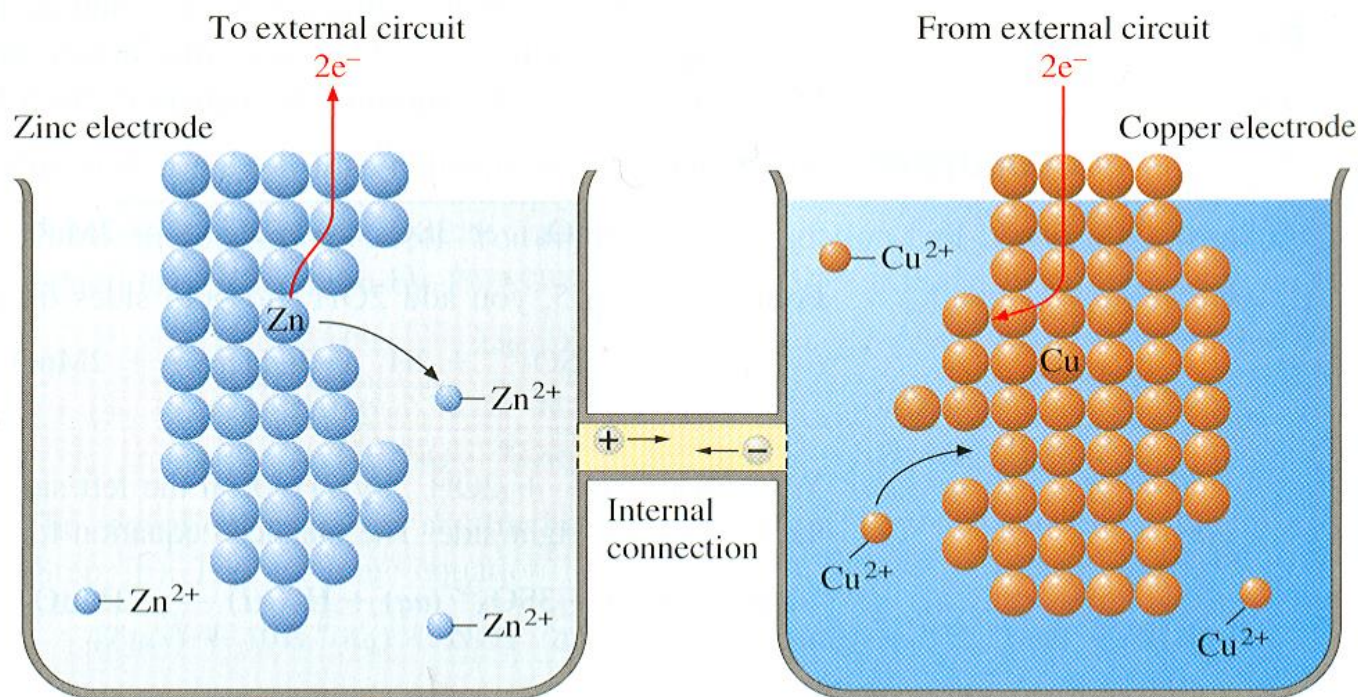


Pharmaceutical Analytical Chemistry II

PCC/PCD 105

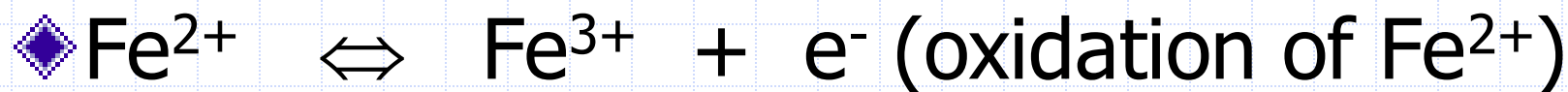
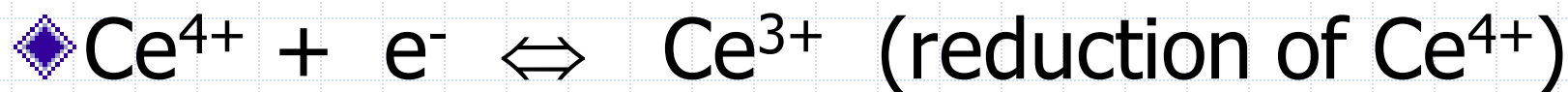
Week 1 lecture

Oxidation Reduction Reactions

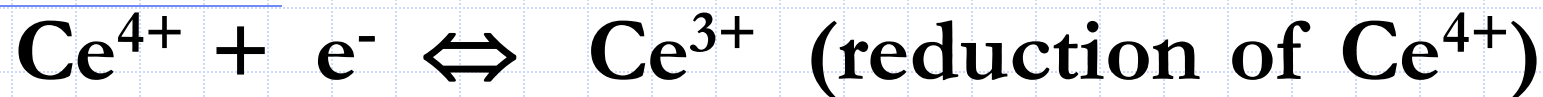


OXIDATION/REDUCTION REACTIONS

◆ An **oxidation reduction reaction** is one in which **electrons are transferred** from one reactant to another involving **change in oxidation number**.



- ◆ **Oxidation:** the loss of electrons by a species, leading to an increase in oxidation number of one or more atoms.
- ◆ **Reduction:** the gain of electrons by a species, leading to a decrease in oxidation number of one or more atoms.
- ◆ **Oxidizing agents (Oxidant):** the species that is *reduced* in a redox reaction.
- ◆ **Reducing agents (Reductant):** the species that is *oxidized* in a redox reaction.



(Overall Redox reaction)



Redox Terminology

PROCESS	$\text{Zn (s)} + 2\text{H}^+(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$	
OXIDATION • One reactant loses electrons • Reducing agent is oxidized • Oxidation number increases	Zinc loses electrons. Zinc is the reducing agent and becomes oxidized. The oxidation number of Zn increases from 0 to +2.	
REDUCTION • Other reactant gains electrons • Oxidizing agent is reduced • Oxidation number decreases	Hydrogen ion gains electrons. Hydrogen ion is the oxidizing agent and becomes reduced. The oxidation number of H^+ decreases from +1 to 0.	

Oxidation Numbers:

Arbitrary numbers assigned to neutral atoms, ions, or molecules to indicate its state of oxidation.

could be positive, negative or a fraction

1. **Elemental substance:** Na, K, H₂, N₂,.....**ZERO**.
2. **Ionic Compounds:** number of electrons gained or lost by elements in forming the ions, e.g., sodium chloride: sodium +1, chlorine -1., NaBr.
3. **Covalent Compounds:** electrons are counted as belonging to the more electronegative of the two atoms. E.g., HCl: chlorine -1, hydrogen +1.
4. **Hydrogen:** **+1**, except in hydrides, **NaH: -1**.
5. **Oxygen:** **-2**, except in peroxides, **H₂O₂, -1**.

Think about it???

◆ **Covalent Compounds:** electrons are counted as belonging to the more electronegative of the two atoms.

◆ Water

◆ Carbon dioxide

◆ Methane

□ Sulphuric acid

□ Sodium tetrathionate.

Balancing Redox Equations

THE HALF-REACTION METHOD

- ◆ This method breaks the overall reaction into its two components – **half-reactions**
- ◆ Each half-reaction is balanced separately and then added
- ◆ Use the following guidelines to help
 1. Write as much of the unbalanced net ionic equation as possible.
 2. Decide which atoms are oxidized and which are reduced – write the two unbalanced half-reactions.
 3. Make equal the number of electrons involved in both half-reactions.
 4. Balance by inspection all atoms in each half-reaction except H and O.
 5. Use the following rules to balance H and O in each half-reaction,

In acidic solution: add H^+ or H_2O only

In basic solution: add OH^- or H_2O only

To balance H

To balance O

In acidic
solution:

Add H^+

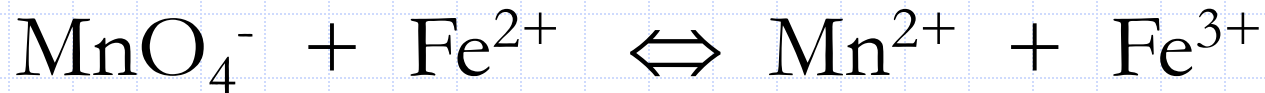
and
then

Add H_2O

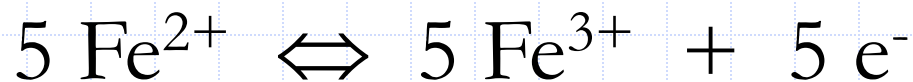
For each O
needed

For each H
needed

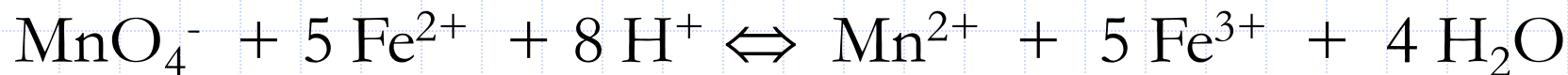
Balance the following net ionic equation in acidic solution



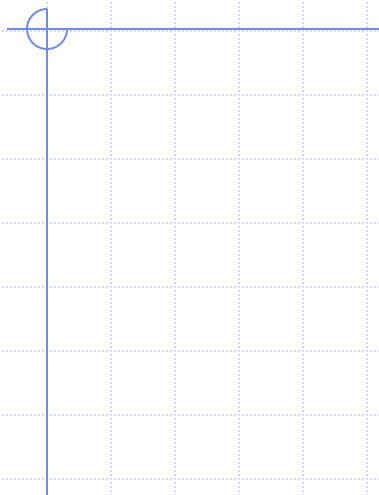
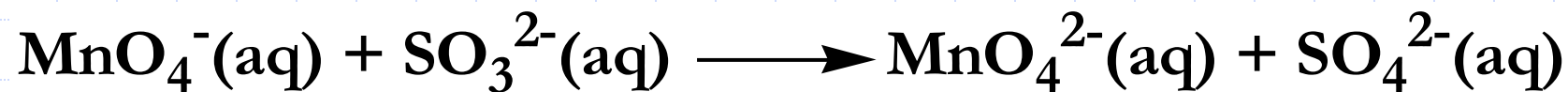
Thus, for the oxidation of Fe^{2+} by MnO_4^- , the half-reactions are:



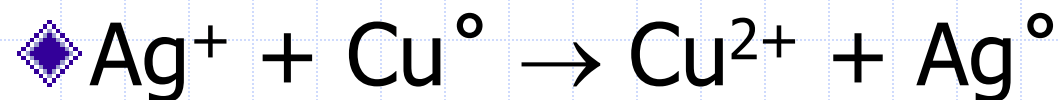
- In the first half-reaction, the net charge on the left side is **$(-1 - 5 + 8) = +2$** , which is the same as that on the right.
- Note also that we have **multiplied** the $\text{Fe}^{2+}/\text{Fe}^{3+}$ half-reaction by **5** so that the number of electrons gained by MnO_4^- equals the number lost by Fe^{2+} .
- A balanced net ionic equation for the overall reaction is then obtained by adding the two half-reactions:



Balance the following net ionic equation in basic solution.



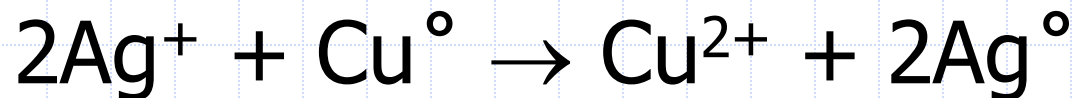
Balance the following equation:



First half reaction: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$

Second half reaction: $2(\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag})$

(Overall balanced equation):



Oxidation/Reduction Reactions in:

1- Redox titrations (In solutions, or **direct contact between reactants**)

e.g. a strip of copper in silver nitrate solution

$2(\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag})$, silver ions are reduced to silver metal, coating the strip of copper.

$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$, Cu metal is oxidized to cupric ions, blue solution.

By time.. The formed Ag metal is deposited on the Cu surface and so reaction stops

2- Electrochemistry (in Electrochemical cells, reactants are separated)

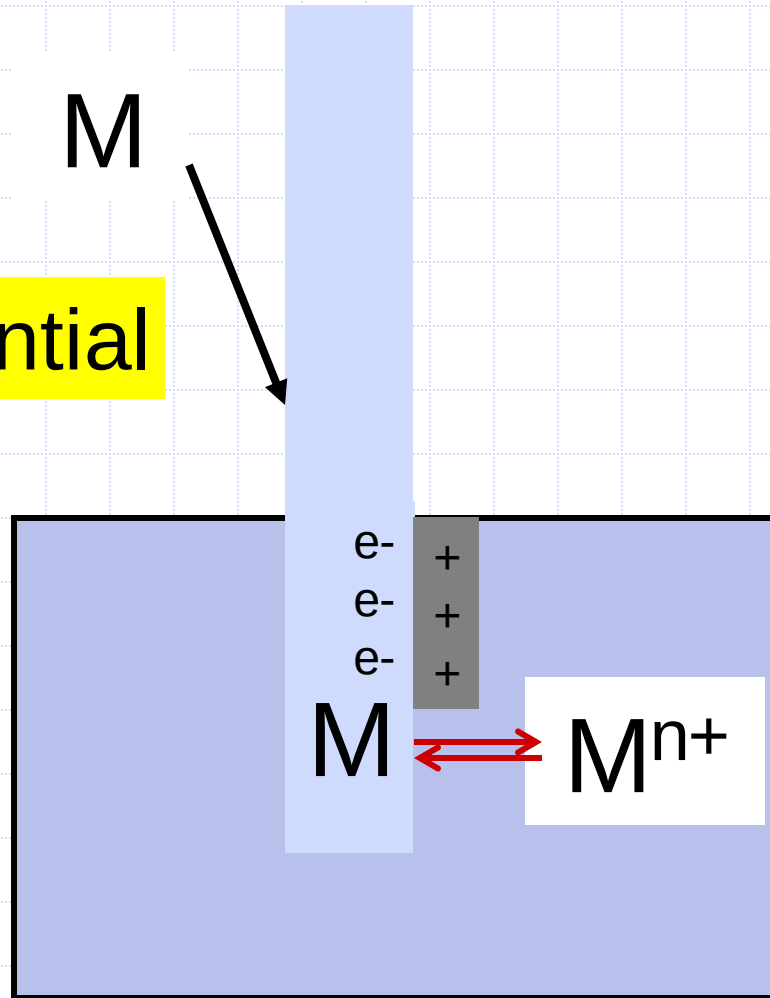
ELECTROCHEMISTRY

Oxidation/Reduction Reactions in Electrochemical Cells

- ◆ All electrochemical reactions involve transfer of electrons and are *redox* reactions.
- ◆ Electrochemical reactions take place in ***electrochemical cell*** (an apparatus that allows a reaction to occur through an external conductor).
- ◆ **No direct contact is involved** (i.e. the oxidizing agent and the reducing agent are physically separated from one another.)

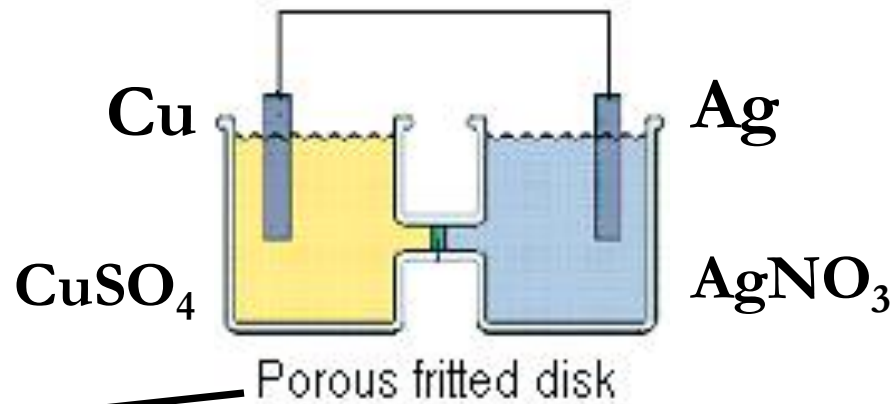
Electrode

Electrode potential



ELECTROCHEMICAL CELLS

- ◆ An electrochemical cell consists of two conductors called *electrodes*, each of which is immersed in an electrolyte solution.



Porous fritted disk

Physical separation of the two halves of cell to **avoid direct Rx** between reactants & allow ions movement

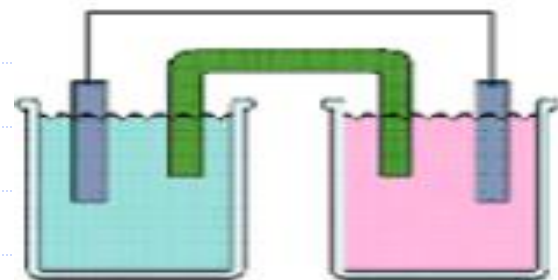
But due to difference in rate migration of **migration "mobility"** of ions across the **interface** (porous fritted disk) → **liquid Junction potential**

Potential that arise at the interface between two electrolyte solutions of different composition.

Problem: contributes to the **total cell potential****interference**

◆ **Solution: Salt bridge** → saturated soln. of **KCl** (ionic soln.) that isolates the two halves of the cell.

◆ as K^+ and Cl^- have the same mobility



Salt bridge

◆ **Conduction of electricity** from one electrolyte solution to the other then occurs by **migration of potassium ions from the bridge in one direction and chloride ions in the other.**

◆ In this way, direct contact between copper metal and silver ions is prevented while electrical continuity is maintained and junction potential is eliminated.

Function of salt bridge

- ◆ Electric current enters and exits the cell by electrodes - **electrodes** are surfaces upon which oxidation or reduction half-reactions occur
- ◆ Two kinds of electrodes:
 1. **Cathode**: - electrode at which **reduction** occurs (electrons are gained by a species)
 2. **Anode**: - electrode at which **oxidation** occurs (as electrons are lost by some species)

- ◆ A unique aspect of oxidation/reduction reactions is that the transfer of electrons can often be brought about in an **electrochemical cell** in which

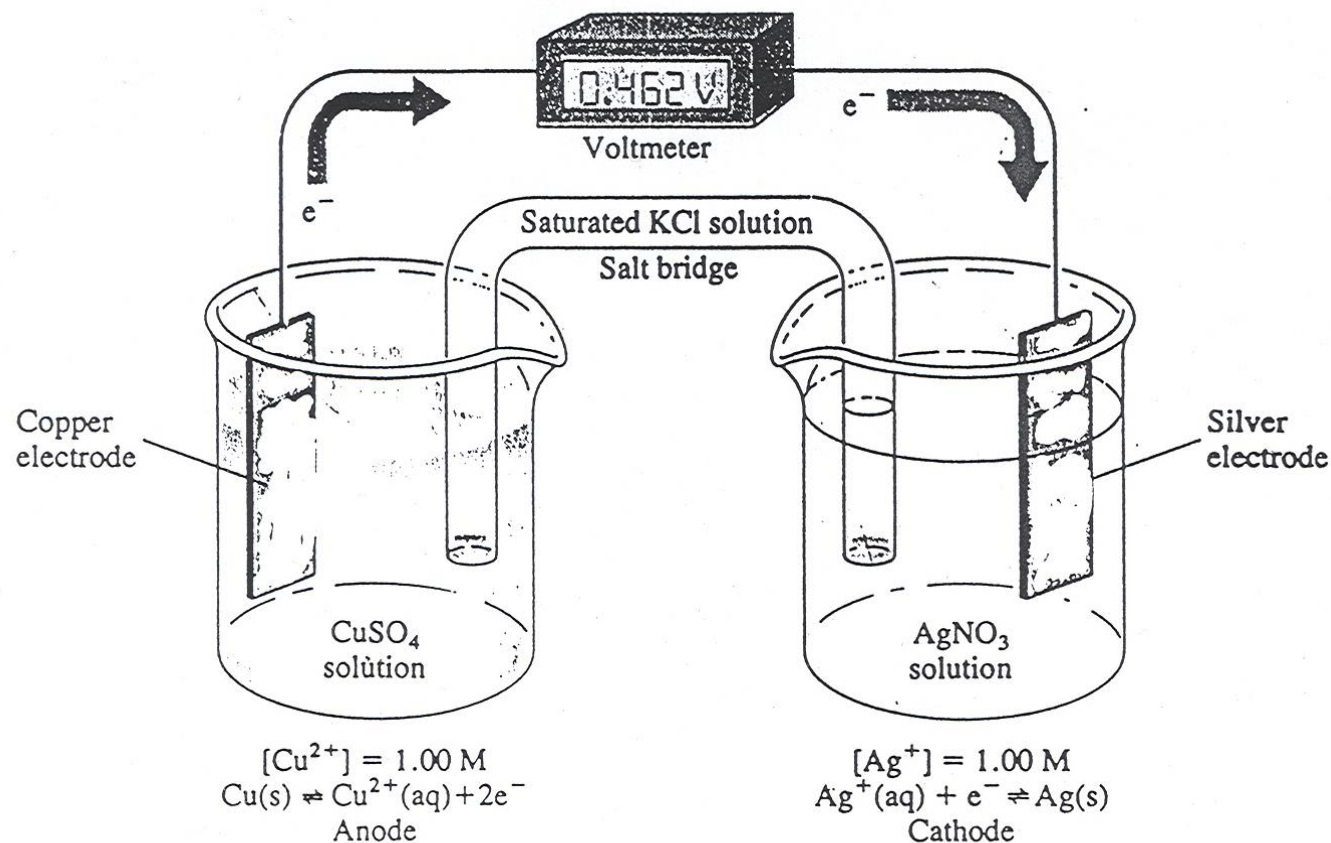


Figure 14-1a
A galvanic cell.

Figure-1(a)

The Silver-Copper cell

Figure 1 (a)- page 20

- Composed of two half-cells:

1. A strip of copper immersed in 1 M CuSO_4
2. A strip of silver immersed in 1 M AgNO_3

- Experimentally we see:

- : - Initial voltage is 0.46 volts
- : - The mass of the copper electrode decreases
- : - The mass of the silver electrode increases
- : - $[\text{Cu}^{2+}]$ increases and $[\text{Ag}^+]$ decreases



ELECTROCHEMICAL CELLS

Two types:

1. **Electrolytic cells:** - these are cells in which an external electrical source forces a *nonspontaneous* reaction to occur
2. **Voltaic cells:** - also called **galvanic cells**. In these cells *spontaneous* chemical reactions *generate electrical energy* and supply it to an external circuit

VOLTAIC (galvanic) CELLS

- Cells in which *spontaneous* reactions produces electrical energy

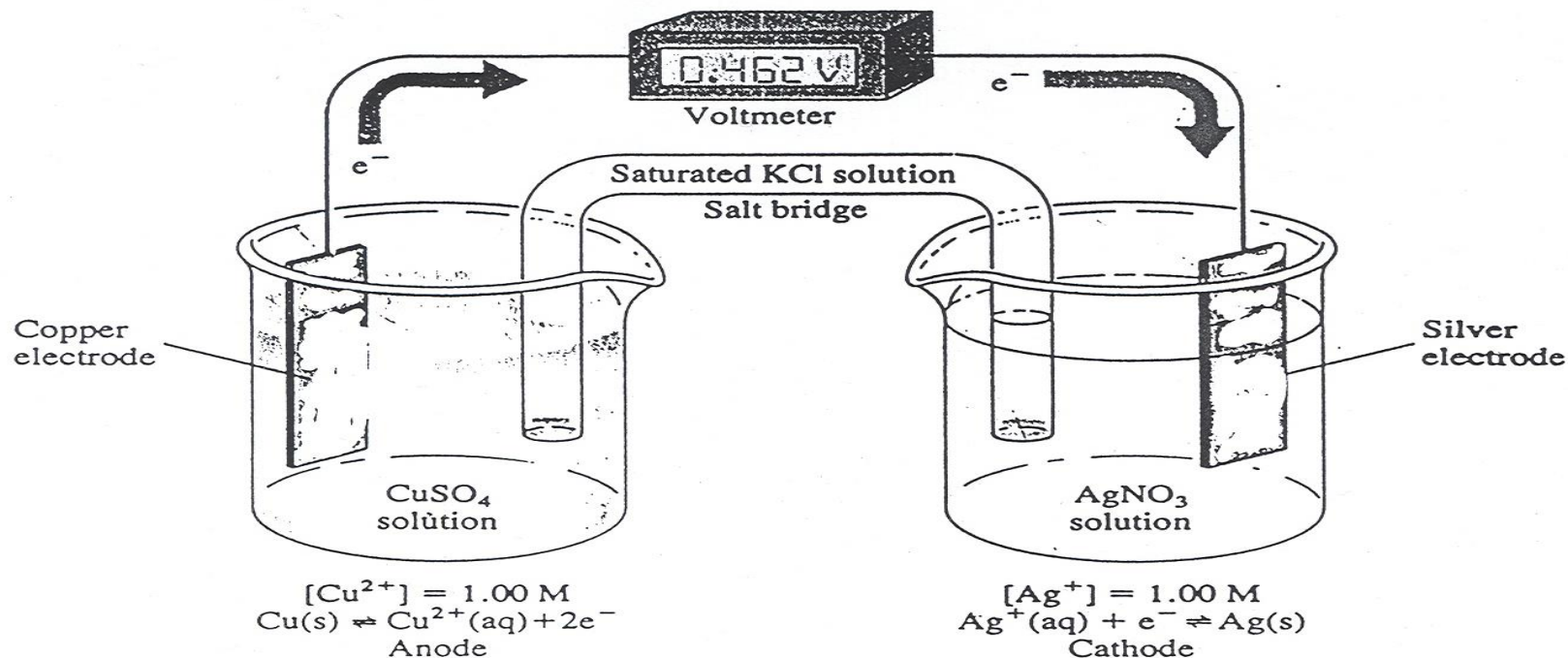


Figure-1(a)



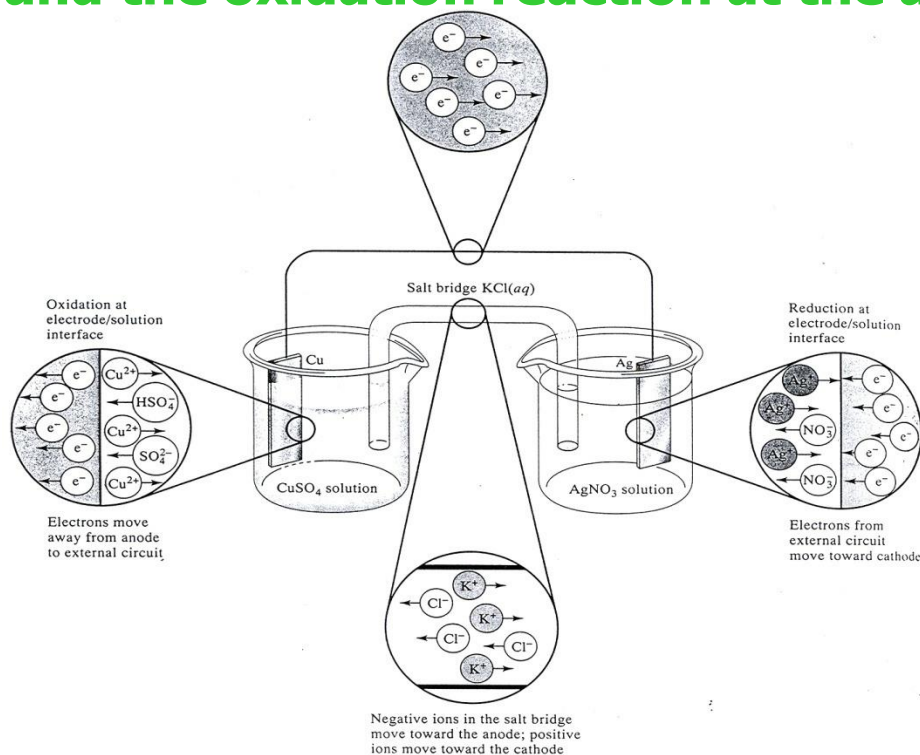
Examples of galvanic cells

- ◆ Dry cells
- ◆ Lead acid battery

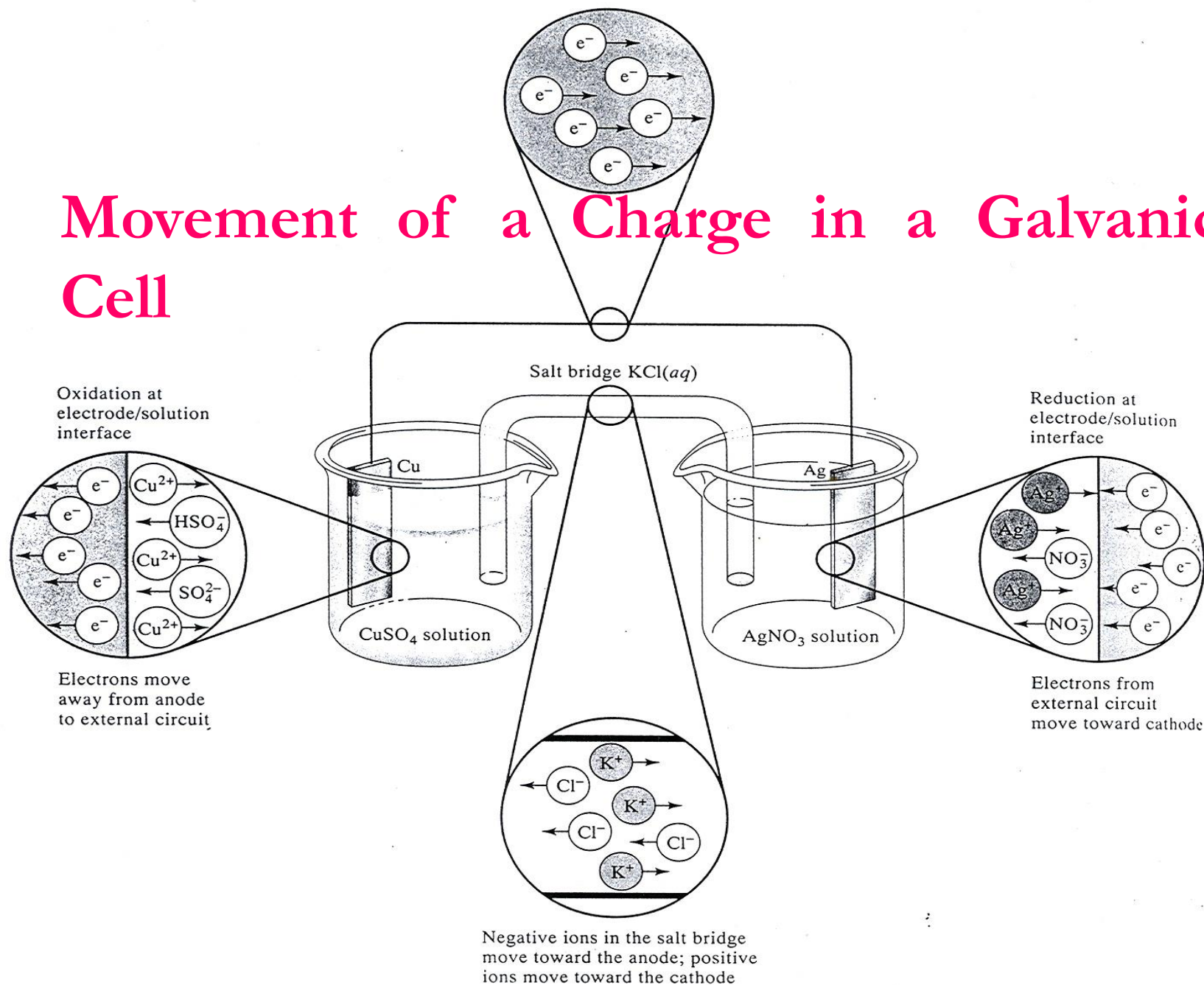
Current in Electrochemical Cells

Electricity is transported by three mechanisms:

- 1- **Electrons** carry electricity within the **electrodes** as well as through the external conductor.
- 2- **Anions and cations** carry electricity within the cell.
- 3- The **ionic conduction** of the solution is coupled to the **electronic conduction** in the electrodes by the **reduction reaction at the cathode** and the **oxidation reaction at the anode**.



Movement of a Charge in a Galvanic Cell



Electrolytic Cells

- ◆ The silver electrode is connected to the positive terminal of the source and becomes the anode of the cell. The copper electrode is connected to the negative terminal and is now the cathode. The silver electrode is connected to the positive terminal and is now the anode. The cell reaction of the electrolytic cell is the reverse of that of the galvanic cell. That is,

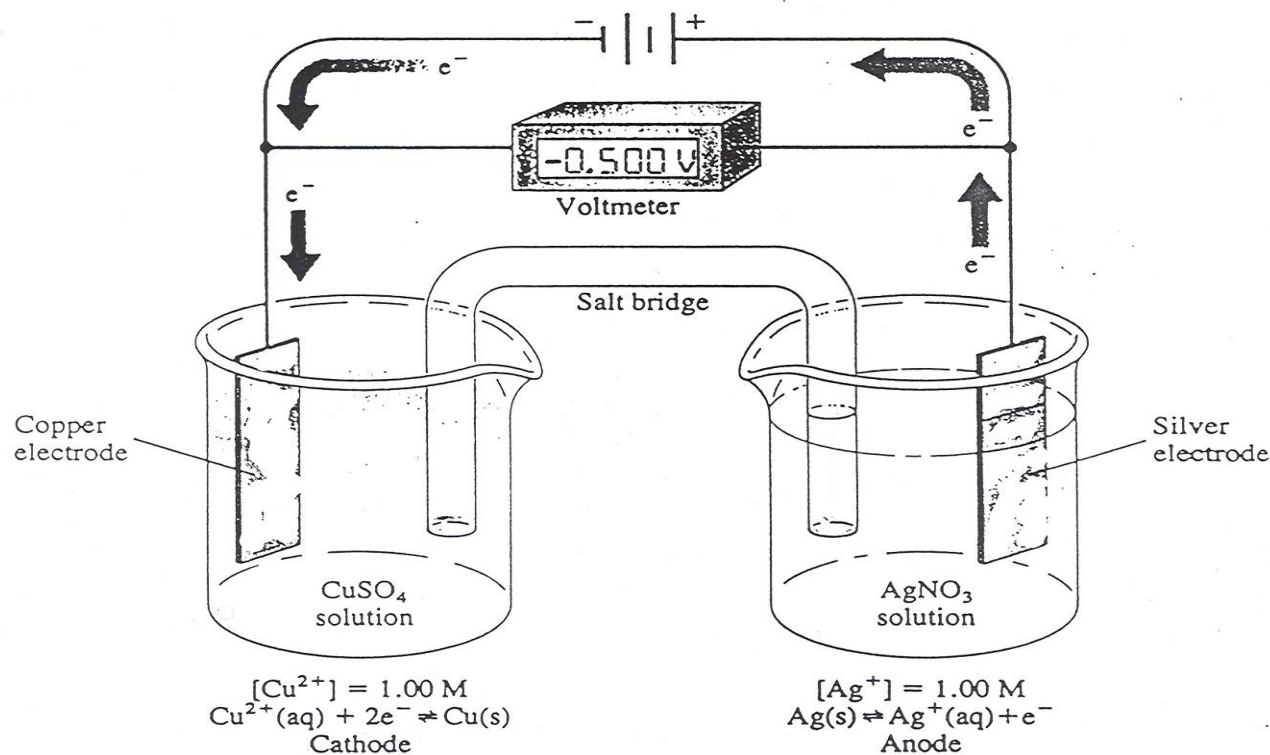
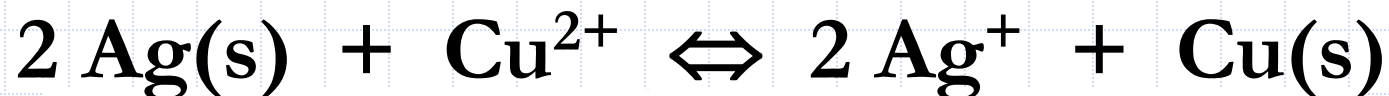


Figure-1(b)

Reversible & Irreversible Cells

- ◆ The cell in Figure-1 is an example of a *reversible cell*, that is, one in which the direction of the electrochemical reaction is reversed when the direction of electron flow is changed.
- ◆ In an *irreversible cell*, changing the direction of current causes entirely different half-reactions to occur at one or both electrodes.

Reversible Cell

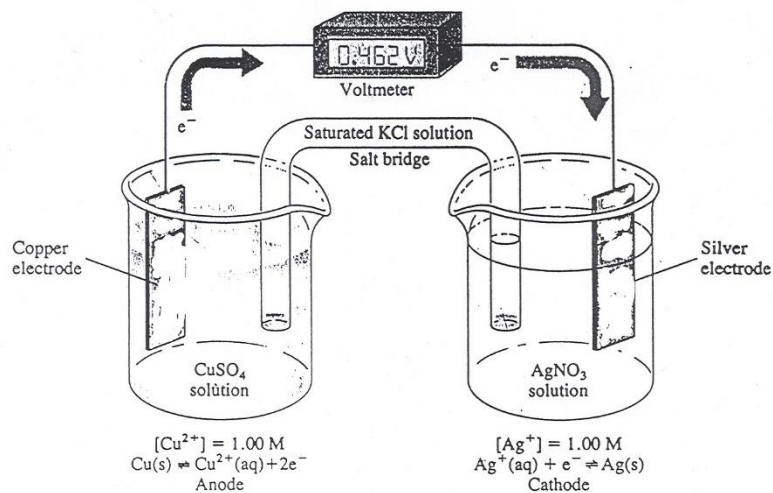


Figure 14-1a
A galvanic cell.

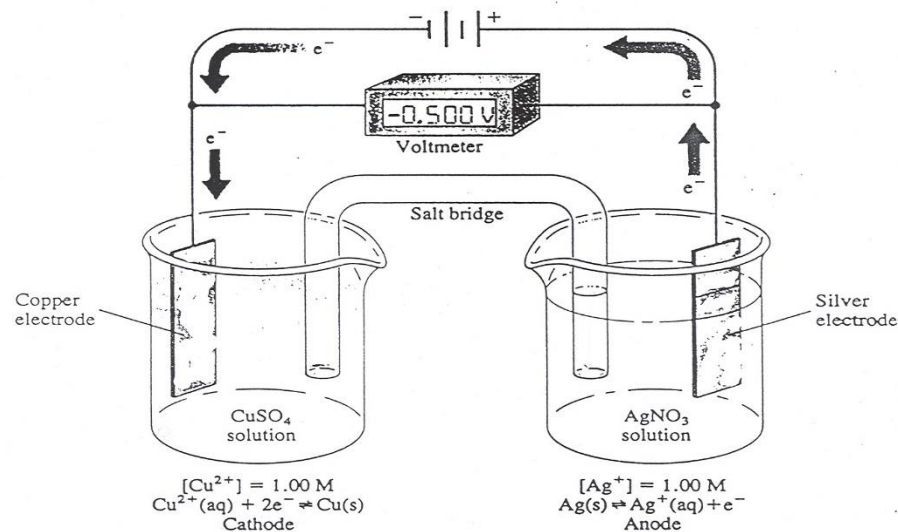
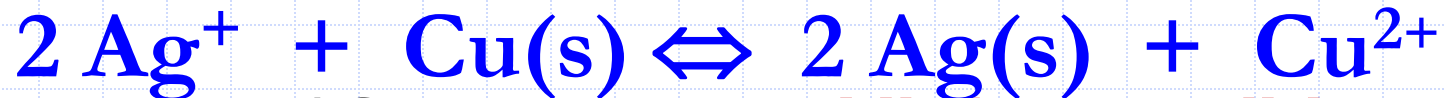


Figure 14-1b
An electrolytic cell.

ELECTRODE POTENTIALS

- ◆ **Cell potential** is **potential difference that develops** between the **cathode and the anode** of **the cell** in Figure-1(a) & it is a measure of the tendency for the reaction:



to proceed **from** a **non-equilibrium condition** to an **equilibrium state**.

- ◆ Thus, as shown in following figure, when the copper and silver ion activities in the cell are **1.00 M**, a potential of **0.462 V develops**, which shows that this reaction is far from equilibrium. As the reaction proceeds, this **potential becomes smaller** until **at equilibrium** the **meter reads 0.00 V**. As shown in the figure, the **copper** ion **equilibrium concentration** is then just slightly less than **1.5 M** and the **silver** ion concentration is **$1.9 \times 10^{-8} \text{ M}$** .

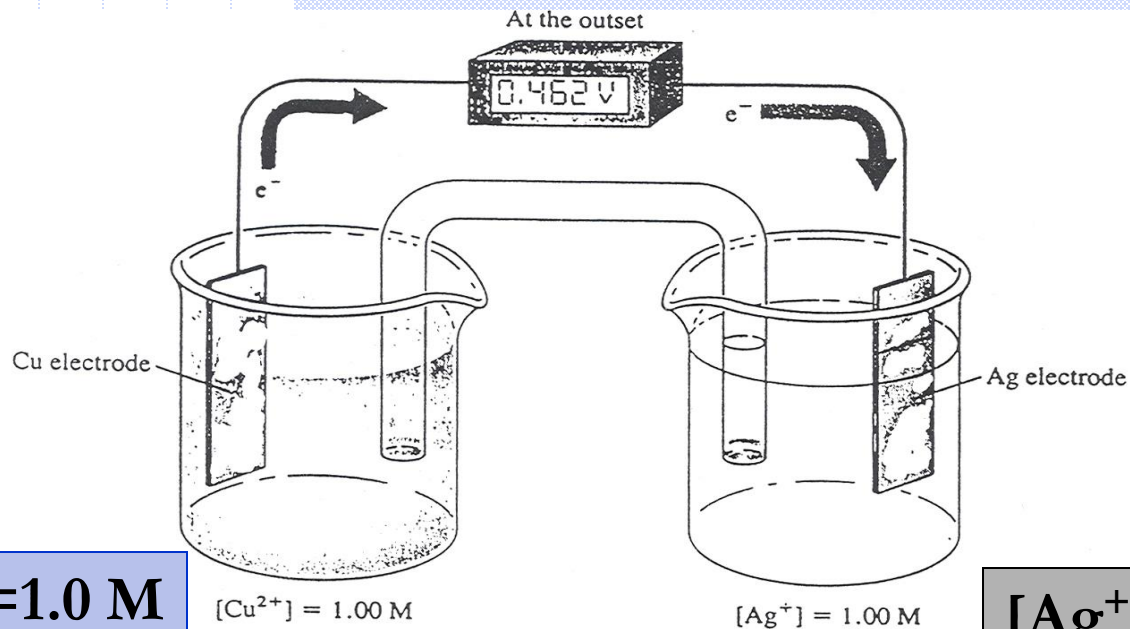
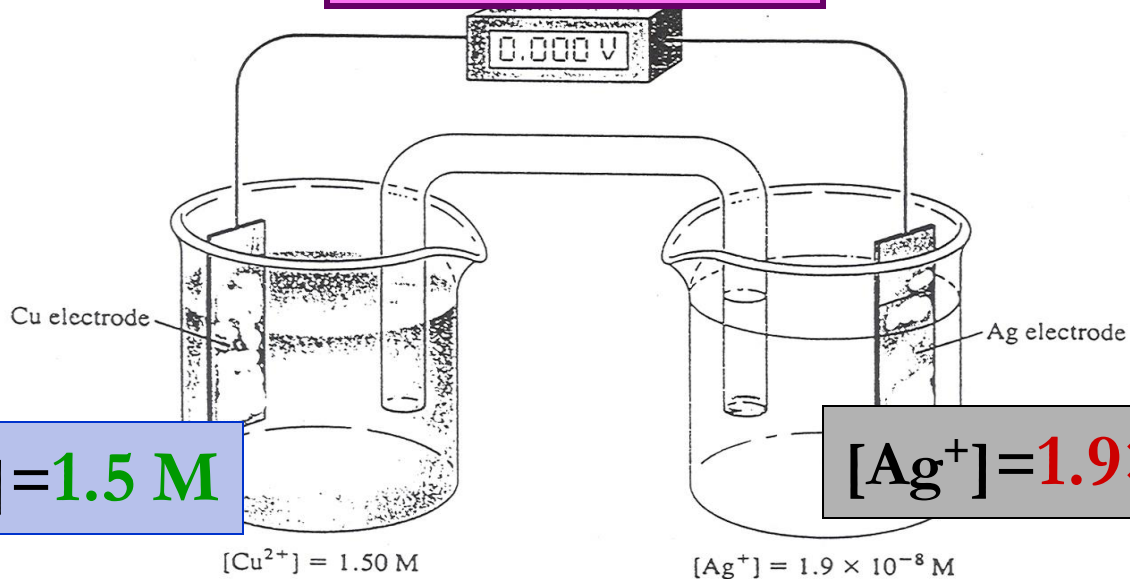


Figure 3: Change in Cell Potential After Passage of Current

AT EQUILIBRIUM



- ◆ When zero voltage is reached, the concentrations of Cu(II) and Ag(I) ions will have values that satisfy the equilibrium-constant
- ◆ . At this point, no further net flow of electrons will occur.

$$K_{eq} = \frac{[Cu^{2+}]}{[Ag^+]^2} = 4.1 \times 10^{15} \quad (4)$$

- ◆ The **potential of cells** such as those shown in the **difference between two *half-cells***, or ***single-electrode potentials***, one associated with the half-reaction at the silver cathode (E_{cathode}) and the other associated with the half-reaction at the copper anode (E_{anode}).
- ◆ Although we **cannot determine absolute potentials of electrodes**, we **can** readily determine ***relative electrode potentials***. For example, if we replace the copper anode in Figure-1(a) with a cadmium electrode immersed in a cadmium sulfate solution, the voltmeter reads about 0.7 V greater than the original cell.

Define:

- ◆ Liquid junction potential
- ◆ Electrode potential
- ◆ Relative electrode potential
- ◆ Cell potential